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ENGINEERING



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Strong and Weak Entities in ER Modeling: Concepts and Real-World Applications

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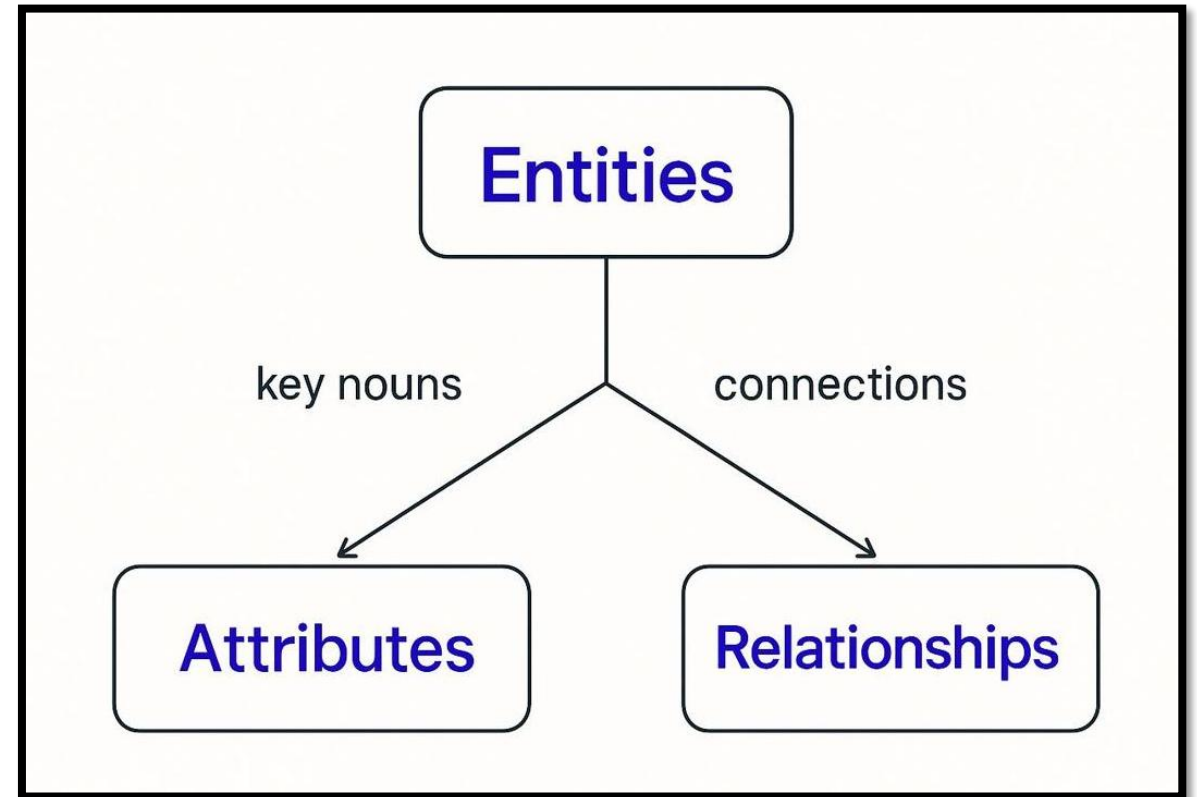
INTRODUCTION

Entity-Relationship Model is used for the designing of databases

- Represents real-world objects as **entities**
- Entities have **attributes** and relationships

Two important types:

- Strong Entity
- Weak Entity

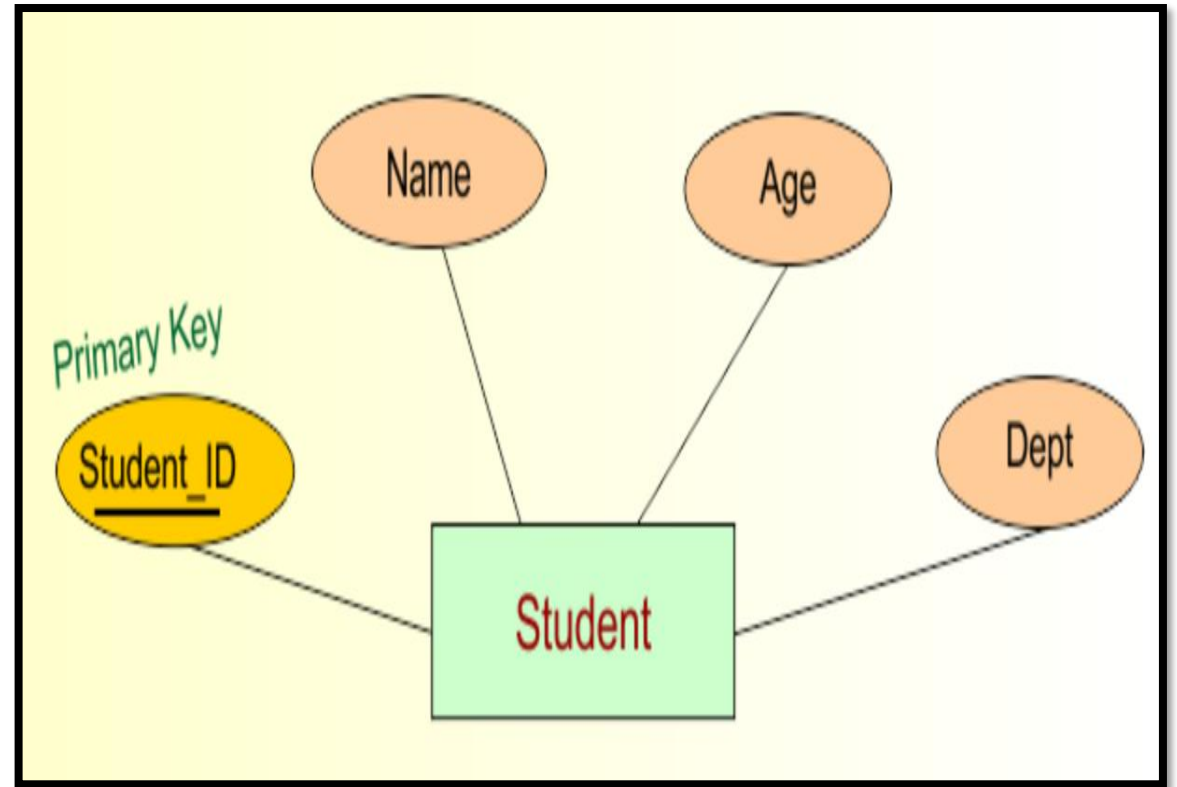


STRONG ENTITY

- A **Strong Entity** has its own primary key
- It can exist independently
- It does not depend on other entities
- Represented by a **single rectangle**

Example:

Student (Student_ID, Name, Dept)



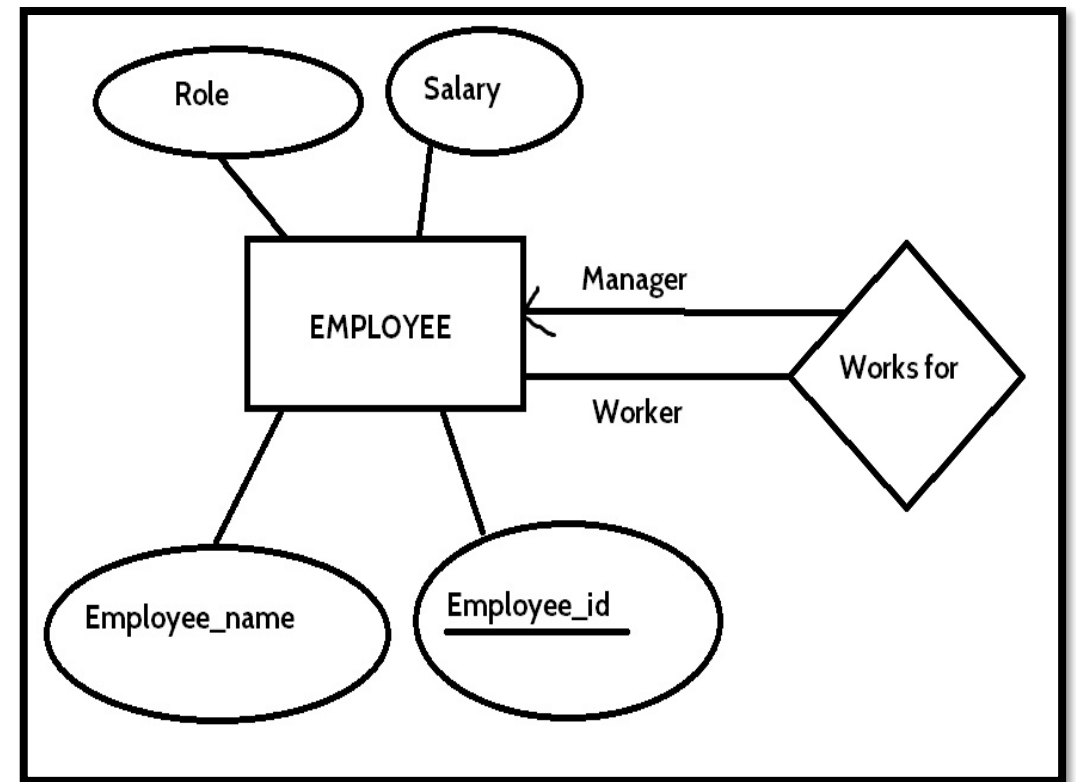
Characteristics of Strong Entity

- It has a **primary key**
- Independent existence
- Can be uniquely identified
- Participates in relationships

Real-world Example:

Employee in a company

Each employee has a unique Employee_ID

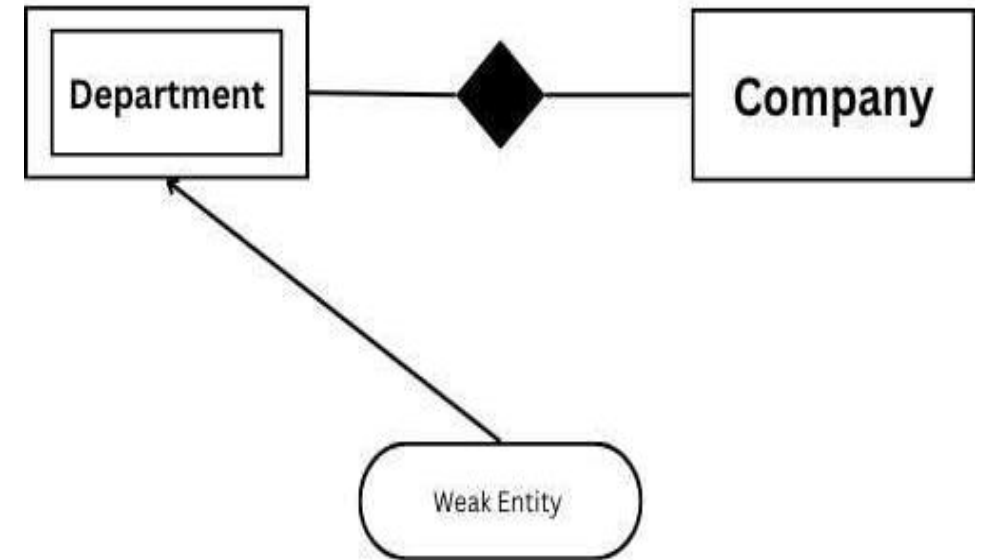


WEAK ENTITY

- A **Weak Entity** does NOT have a primary key
- Depends on a strong entity
- Identified using a **partial key + owner entity key**
- Represented by a **double rectangle**

Example:

Dependent (Department_Name, Company)



Characteristics of Weak Entity

- It has no primary key
- Depends on strong entity
- Has a **partial key (discriminator)**
- Requires identifying relationship



Real-world Example:

Bank Account Transaction depends on Account

Strong Vs Weak Entity

Feature	Strong Entity	Weak Entity
 Definition	An entity that can exist independently.	An entity that cannot exist independently; it depends on another entity.
 Primary Key	Has its own primary key.	Does not have a full primary key.
 Key Type	Primary Key	Partial Key (Discriminator)
 Existence	Exists independently of other entities.	Exists only with the strong (owner) entity.
 Relationship	Not dependent on any other entity.	Dependent on strong entity through an identifying relationship.
 Representation in ER Diagram		
 Identifying Relationship		
 Example	Student, Employee, Department, Course	Order Item, Dependent, Marks, Payment

Real-World Example

Strong Entity: Student

- Has **Primary Key (Student_ID)** And Exists independently

Weak Entity: Exam_Result

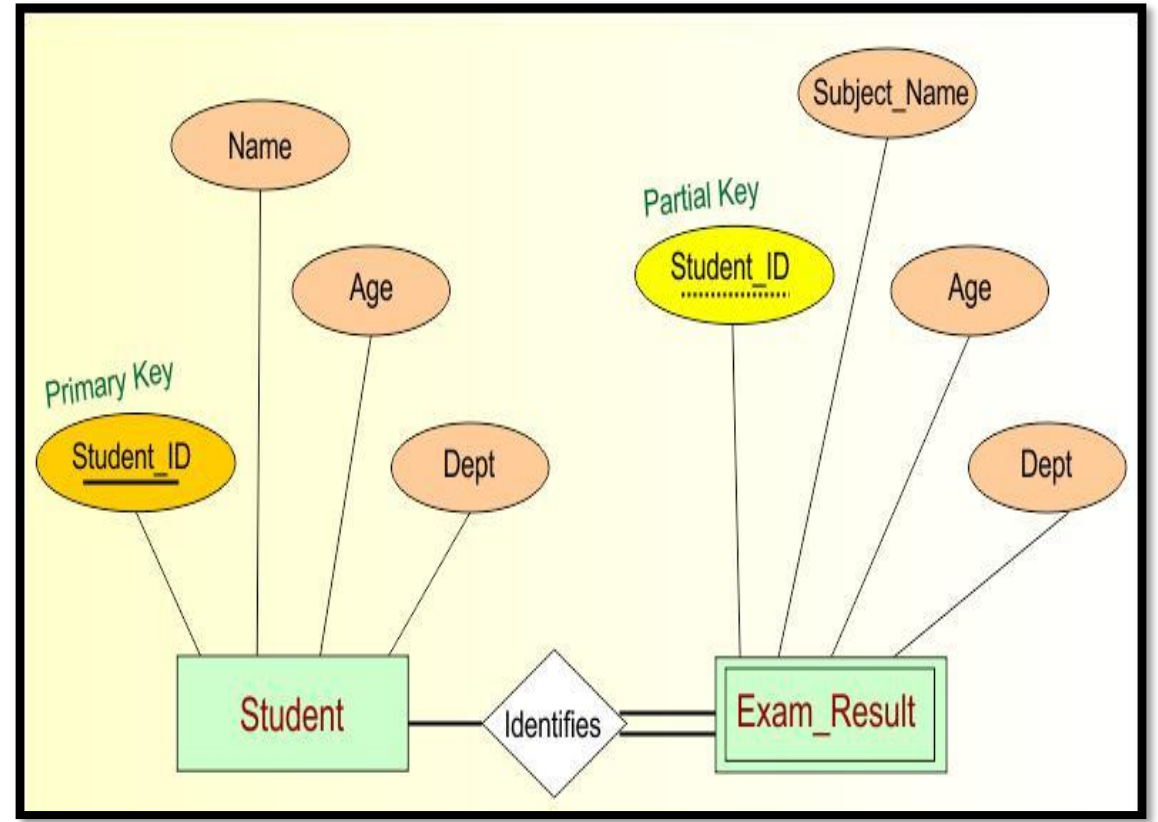
- Depends on **Student** and Has **Partial Key (Subject_Name)** and also Identified by **Student_ID + Subject_Name**

Application

Used in **college databases** for storing marks

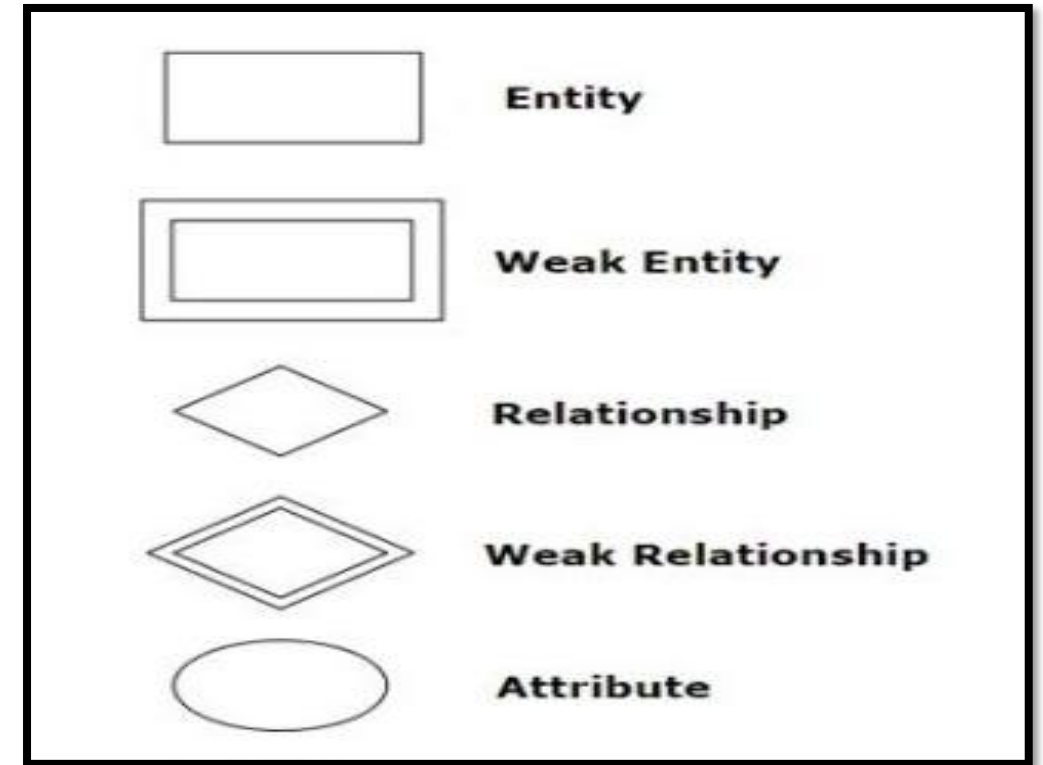
Student → Independent

Exam_Result → Dependent



ER Diagram Representation

- Strong Entity → Rectangle
- Weak Entity → Double Rectangle
- Identifying Relationship → Double Diamond
- Partial Key → Dashed underline



CONCLUSION

- Strong entities are **independent** and self-identified
- Weak entities are **dependent** and need a strong entity
- Both are important for real-world database design
- Helps in building efficient and meaningful ER diagrams



*Thank
You*